Learning-Guided Tensor Nuclear Norm for Hyperspectral Image Inpainting

Yunshan Li, **WENWU GONG***, Chao, Wang and Lili Yang

Department of Statistics and Data Science
Southern University of Science and Technology, Shenzhen, China

*Corresponding and co-first author

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Outline

1. Introduction and Motivation
2. Evolution of TNN Methods
3. Problem I: Weighted Nuclear Norm
4. Problem II: Direction Inflexibility
5. Conclusion and Future Work

Hyperspectral Images Enable Critical Applications

Hyperspectral Images (HSIs) encode detailed spatial–spectral information through hundreds of contiguous bands, enabling reliable analysis in mission-critical applications such as environmental monitoring and urban sensing.

Key Applications:

- **Environmental Monitoring**: vegetation health assessment and pollution detection
- **Urban Planning**: land cover classification and urban expansion monitoring
- **Precision Agriculture**: crop health monitoring, soil assessment, pest management

Critical Challenge: real-world HSI data frequently suffer from degradation due to sensor malfunctions, atmospheric interference, and transmission errors.

Need: effective HSI inpainting techniques to reconstruct hyperspectral data.

HSI Inpainting as Low-rank Tensor Completion

Problem Formulation: an HSI is naturally represented as a three-way tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, where the first two modes correspond to spatial dimensions and the third to the spectral dimension.

Classical Low-rank Formulation:

$$\min_{\mathcal{X}} \text{rank}(\mathcal{X}) + \lambda \mathcal{L}(\mathcal{X}, \mathcal{O})$$

Tensor Nuclear Norm (TNN) as convex surrogate for rank minimization:

$$\min_{\mathcal{X}} \sum_{k} \|\hat{\mathcal{X}}^{(k)}\|_* + \lambda \mathcal{L}(\mathcal{X}, \mathcal{O}) \quad \text{s.t.} \quad \mathcal{X} = g(\hat{\mathcal{X}})$$

where $g(\cdot)$ is a transform (e.g., inverse DFT) mapping latent tensor $\hat{\mathcal{X}}$ to image $\mathcal{X}$.

Key Idea: exploit the inherent low-rank structure by imposing a nuclear norm regularization on the latent tensor.

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Evolution of TNN Methods

Traditional TNN (Fixed Transforms)

Methods: TNN, 3DTNN, WTNN
Limitation: transforms lack of adaptivity to data characteristics; nuclear norm ignores channel differences and inter-channel relationships

Single-direction TNN

Methods: S2NTNN (1D fully connected), CoNoT (1D coupled nonlinear)
Feature: learnable transforms, primarily along spectral mode
Limitation: single direction, fails to learn spatial–spectral correlations across all tensor modes

Our Contributions

CA-TNN: channel-aware weighting for fine-grained inter-channel relationships
3DeepRep: multi-directional deep representations for enhanced exploitation of low-rank tensor structures

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Problem I - Weighted Nuclear Norm

Core Problem

Existing WTNN methods ignore distinct characteristics of latent tensor channels and their inter-channel interactions.

$$\|\mathcal{X}\|_{\mathbf{W},*} = \sum_{k=1}^{\widetilde{n_3}} \sum_{i=1}^{\min(n_1,n_2)} \mathbf{W}(i,k)\sigma_i(\widehat{\mathcal{X}}^{(k)}),$$

where $\mathbf{W} \in \mathbb{R}^{\min(n_1,n_2)\times\widetilde{n_3}}$ is a nonnegative weight matrix, and $\sigma_i(\widehat{\mathcal{X}}^{(k)})$ denotes the i-th singular value of the k-th frontal slice of $\widehat{\mathcal{X}}$.

Two Classes of WTNN:

- Given $\mathbf{W}(i,1) = \mathbf{W}(i,2) = \cdots \mathbf{W}(i,\widetilde{n_3})$, ignore inter-channel differences
- Learnable $\mathbf{W}(i,k)$, overlook channel characteristics

Consequence: suboptimal restoration performance for HSI data with complex properties.

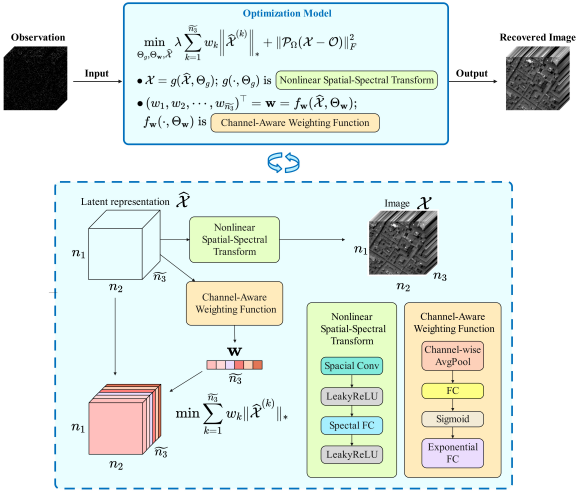
CA-TNN Model

Insights: explicitly model inter-channel relationships through attention mechanism.

Key Components:

1. **Learnable nonlinear spatial-spectral transform** $g(\cdot, \Theta_g)$
2. **Channel-aware weighting function** $f_{\mathbf{w}}(\cdot, \Theta_{\mathbf{w}})$
3. **Channel-aware TNN**:

$$\|\mathcal{X}\|_{\text{CA-TNN},*} = \sum_{k=1}^{\tilde{n}_3} w_k \|\hat{\mathcal{X}}^{(k)}\|_*$$



Channel-Aware Weighting Function

Squeezing Step: extract channel characteristics via global average pooling

$$z_k = \frac{1}{n_1 n_2} \sum_{l=1}^{n_1} \sum_{j=1}^{n_2} \hat{\mathcal{X}}^{(k)}(l, j), \quad f_{sq}(\hat{\mathcal{X}}) = (z_1, \ldots, z_{\tilde{n}_3})^\top$$

Excitation Step: model inter-channel relationships via exponential FC layer

$$\mathbf{w} = e^{\mathbf{b}_2} + e^{\mathbf{W}_2} \delta(\mathbf{W}_1 f_{sq}(\hat{\mathcal{X}}))$$

- Weight range: $(0, +\infty)$ vs. original SE's $(0, 1)$
- Greater flexibility in channel importance modeling
- Learned weights decrease during optimization

Innovation: inspired by squeeze-and-Excitation (SE) attention mechanism, but with exponential output for enhanced flexibility.

CA-TNN Optimization

Optimization Model:

$$\min_{\hat{\mathcal{X}},\Theta_{\mathbf{w}},\Theta_g} \lambda \sum_{k=1}^{\tilde{n}_3} w_k \|\hat{\mathcal{X}}^{(k)}\|_* + \|\mathcal{P}_\Omega(g(\hat{\mathcal{X}},\Theta_g) - \mathcal{O})\|_F^2$$

Learnable Variables:

- $\hat{\mathcal{X}}$: Latent tensor
- $\Theta_{\mathbf{w}}$: Channel weighting function parameters $(\mathbf{W}_1, \mathbf{W}_2, \mathbf{b}_2)$
- Θ_g: Transform parameters (spatial and spectral)

Self-Supervised Learning: Adam optimizer with gradient descent, no ground-truth required

CA-TNN Experimental Results

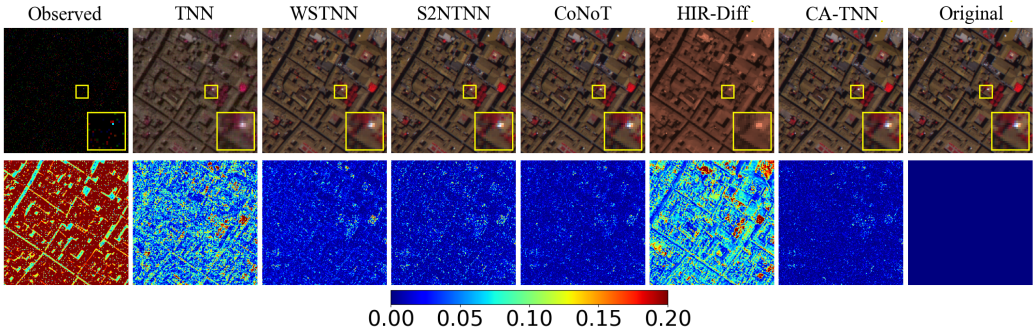


Figure: Recovered images of different methods on Pavia with MR = 0.95. The top row displays false-color images using bands 75 (R), 35 (G), and 5 (B). The bottom row shows the absolute error maps between the ground truth and the recovered images (on band 75).

Finding: CA-TNN's false-color image shows closest color tone to ground truth, error map closest to pure dark blue.

Table: Quantitative Performance: Red and Blue Indicate the Best and Second-Best Values, Respectively.

Dataset	MR	Metric	Observed	TNN	WSTNN	S2NTNN	CoNoT	CA-TNN
Pavia	0.99	PSNR	12.013	12.013	17.552	26.001	29.523	29.815
		SSIM	0.004	0.004	0.243	0.755	0.885	0.895
		SAM	2.207	2.207	0.178	0.090	0.075	0.074
	0.95	PSNR	12.192	28.879	37.296	37.932	40.456	42.023
		SSIM	0.018	0.860	0.981	0.981	0.989	0.993
		SAM	1.381	0.126	0.034	0.033	0.026	0.022
	0.90	PSNR	12.426	32.637	45.952	46.416	48.980	51.140
		SSIM	0.035	0.930	0.997	0.997	0.998	0.999
		SAM	1.255	0.097	0.013	0.014	0.012	0.008
DCWall	0.99	PSNR	25.615	25.615	31.825	40.718	43.150	44.161
		SSIM	0.266	0.266	0.604	0.908	0.940	0.958
		SAM	1.725	1.725	0.295	0.105	0.090	0.072
	0.95	PSNR	25.794	33.314	48.903	50.505	53.151	55.403
		SSIM	0.289	0.811	0.986	0.990	0.994	0.997
		SAM	1.355	0.155	0.046	0.032	0.029	0.023
	0.90	PSNR	25.615	36.826	56.240	54.997	57.666	60.694
		SSIM	0.241	0.902	0.998	0.998	0.998	0.999
		SAM	1.725	0.106	0.020	0.018	0.019	0.014

Finding: consistent superiority over WSTNN, **CoNoT**, and **S2NTNN**, across all metrics and scenarios

CA-TNN Ablation Study

We compare our proposed exponential SE (ExpSE) function with three other functions:

	Method	PSNR	SSIM	SAM
Weighting Scheme	WoW	40.661	0.990	0.026
	ChannelW	42.023	0.993	0.022
	SingularW	39.270	0.986	0.032
	ChannelW + SingularW	41.173	0.991	0.023
Channel Weighting Function	SoftmaxSE	40.878	0.990	0.025
	SigmoidSE	37.443	0.977	0.040
	ExpSE	42.023	0.993	0.022
	N-ExpSE	40.127	0.989	0.028

Finding: ExpSE with normalization achieves superior performance.

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Problem II - Direction Inflexibility

Core Problem

Existing transform-based TNN methods (including CA-TNN) typically apply the transform along a single tensor mode (most often the spectral mode), thereby neglecting low-rank structure in the remaining modes.

$$\|\mathcal{X}\|_{\text{3DTNN}} = \alpha_i \sum_{k=1}^{\widetilde{n_i}} \left\| \left(\widehat{\mathcal{X}}_i \right)_i^{(k)} \right\|_*$$

Existing 3D Extensions:

- 3DTNN/WSTNN: fixed multi-mode transforms, lack adaptivity
- t-CTV: handcrafted priors, manual hyperparameter tuning
- 3DS2NTNN: separate mode-specific networks, manual fusion

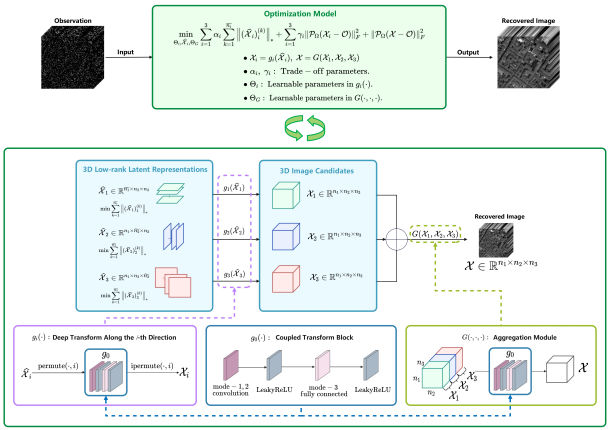
Consequence: fail to capture cross-directional correlations that are critical for HSI recovery.

3DeepRep Model

Insight: enforce low-rankness along all three modes by minimizing nuclear norms of mode-i frontal slices in latent spaces.

Key Components:

1. **Three latent tensors** $\{\hat{\mathcal{X}}_i\}_{i=1}^3$, each for one directional representation
2. **Multi-directional deep transforms** $\{g_i(\cdot)\}_{i=1}^3$ that map latent tensors to image candidates $\{\mathcal{X}_i\}_{i=1}^3$
3. **Learnable aggregation module** $G(\mathcal{X}_1, \mathcal{X}_2, \mathcal{X}_3)$ fuses directional reconstructions



Coupled Transform Block (CTB)

The **Coupled Transform Block (CTB)** serves as the fundamental module for constructing directional deep transforms.

$$g_0(\cdot) = w_2(w_1(\cdot))$$

which enables learning nonlinear features for low-rank latent tensors.

Two-stage Transform:

- Spatial Transform $w_1(\cdot)$:
  - Applies slice-wise 2D convolution (3×3 kernel) to each frontal slice
  - Captures local spatial correlations within slices
  - Followed by LeakyReLU activation
- Spectral Transform $w_2(\cdot)$:
  - Performs mode-3 tensor-matrix product with learnable weight matrix
  - Models inter-slice correlations along the spectral mode
  - Followed by LeakyReLU activation

Multi-directional Deep Transform

We extend the CTB to all three tensor modes using mode permutation operators.

Directional Deep Transform (for i-th direction, $i = 1, 2, 3$):

$$g_i(\cdot) = \text{ipermute}(g_0(\text{permute}(\cdot, i)), i), \quad \mathcal{X}_i = g_i(\hat{\mathcal{X}}_i)$$

- Permute($\cdot$, i): rearranges tensor so that mode-i becomes the third mode
- CTB $g_0(\cdot)$: applies spatial-spectral transform in the permuted space
- Ipermute($\cdot$, i): restores original tensor orientation

Comprehensive Modeling: each directional transform captures mode-specific correlations:

- $g_1(\cdot)$: Exploits correlations along mode-1 (first spatial dimension)
- $g_2(\cdot)$: Exploits correlations along mode-2 (second spatial dimension)
- $g_3(\cdot)$: Exploits correlations along mode-3 (spectral dimension)

Learnable Aggregation Module

The aggregation module learns optimal fusion weights through gradient-based optimization.

Aggregation Formulation:

$$\mathcal{X} = G(\mathcal{X}_1, \mathcal{X}_2, \mathcal{X}_3) = g_3(\text{concat}(\mathcal{X}_1, \mathcal{X}_2, \mathcal{X}_3))$$

- Concatenation: stacks three directional reconstructions along spectral dimension, creating a tensor with $3n_3$ bands
- Spectral-mode Transform $g_3(\cdot)$: applies CTB to capture inter-band dependencies across directional reconstructions
- Adaptive Fusion: parameters Θ_G learned via back-propagation to automatically balance directional contributions

Advantage over Manual Fusion: unlike traditional ADMM-based frameworks requiring manual tuning of multiple hyperparameters, our learnable module significantly reduces reliance on handcrafted parameters and improves robustness.

3DeepRep Optimization

The 3DeepRep model formulates HSI inpainting as a self-supervised optimization problem.

Optimization Objective:

$$\min_{\Theta_i, \hat{\mathcal{X}}_i, \Theta_G} \sum_{i=1}^{3} \alpha_i \mathcal{L}_i(\Theta_i, \hat{\mathcal{X}}_i) + \|\mathcal{P}_\Omega(G(\mathcal{X}_1, \mathcal{X}_2, \mathcal{X}_3) - \mathcal{O})\|_F^2$$

where

$$\mathcal{L}_i(\Theta_i, \hat{\mathcal{X}}_i) = \sum_{k=1}^{\tilde{n}_i} \|(\hat{\mathcal{X}}_i)_i^{(k)}\|_* + \lambda_i \|\mathcal{P}_\Omega(\mathcal{X}_i - \mathcal{O})\|_F^2$$

Learnable Variables:

- **Transformation weights** $\{\Theta_i\}_{i=1}^3$: parameters of directional deep transforms
- **Aggregation parameter** Θ_G: parameters of fusion module
- **Latent tensors** $\{\hat{\mathcal{X}}_i\}_{i=1}^3$: directly optimized as learnable variables

Fully Gradient-driven: Adam optimizer with self-supervised learning.

Datasets and Evaluation Metrics

Datasets:

- Pavia Centre: 200×200×80 bands
- Pavia University: 256×256×103 bands
- Washington DC Mall: 256×256×191 bands
- **MODIS Level-1B** (Real): Antarctic Peninsula, 256×256×10 bands

Evaluation Metrics:

- PSNR: Spatial quality, higher is better
- SSIM: Structural preservation, higher is better
- SAM: Spectral fidelity, lower is better

Missing Scenarios:

- Case 1: Point missing (0.85-0.99)
- Case 2: Stripe slice missing (0.10-0.50)
- Case 3: Deadline missing (varying widths)
- Case 4: Mixed missing (Case 1+2+3)

Baselines: Model-Guided and Learning-Guided Paradigms

Model-Guided Methods (Pre-defined transforms, explicit optimization):

- **TNN**: classical Fourier-based TNN, 1D transform framework
- **3DTNN (WSTNN)**: 3D extension with predefined multi-mode transforms
- **t-CTV**: low-rankness + 3D total variation, handcrafted priors

Learning-Guided Methods (Deep transforms, gradient-based optimization):

- **S2NTNN**: 1D learnable transform via fully connected networks
- **CoNoT**: coupled nonlinear 1D transform, spatial-spectral learning
- **3DSTNN**: 3D extension of S2NTNN, sensitive to initialization
- **HIR-Diff**: diffusion model + low-rank decomposition

Our Contribution - 3DeepRep: Fully learning-based 3D TNN with three key innovations:

1. Joint learnable transformation and tensor estimation
2. Fully gradient-driven optimization without explicit solver
3. Robust to initialization, stable convergence

3DeepRep Provides Greatest Impact on Challenging Scenarios

Representative Results:

Case	MR	PSNR	SSIM	SAM	vs. 2nd Best
1	0.99	**30.754**	**0.915**	**0.056**	+1.231 dB (CoNoT)
1	0.85	**58.660**	**1.000**	**0.003**	+3.670 dB (CoNoT)
2	0.10	**47.956**	**0.998**	0.011	+4.939 dB (CoNoT)
3	-	**51.052**	**0.999**	**0.007**	+7.553 dB (CoNoT)
4	-	**39.514**	**0.987**	**0.021**	+6.235 dB (CoNoT)

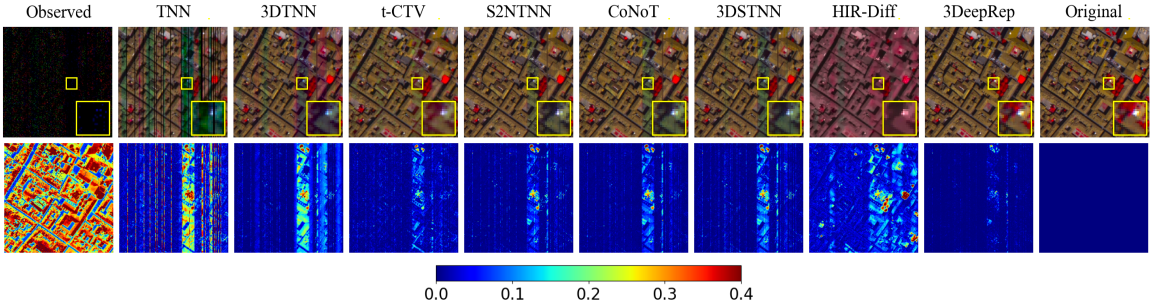
Overall Performance (Average PSNR improvement over 2nd best):

- **Pavia Centre**: +4.7 dB average gain
- **Pavia University**: +2.4 dB average gain
- **Washington DC Mall**: +1.3 dB average gain

Finding: 3D structure provides substantial advantage in complex degradation (Case 3, 4)

Visual Results Demonstrate Superior Spatial-Spectral Recovery

Case 4 (Mixed Missing) Visual Comparison:



- **TNN, 3DTNN, S2NTNN, CoNoT, 3DSTNN**: noticeable stripe noise
- **t-CTV**: blurred details, over-smoothing
- **HIR-Diff**: color tone deviation, spectral distortion

Finding: multi-directional transforms capture both spatial boundaries and spectral consistency

Ablation Study

Experiment: compare 3D transform vs. 1D transforms (mode-1, mode-2, mode-3 only)

Results (Case 4, Pavia Centre):

Configuration	PSNR	SSIM	SAM	Gain vs. Best 1D
Mode-1 only	30.838	0.925	0.077	-
Mode-2 only	21.907	0.647	0.266	-
Mode-3 only	33.939	0.961	0.036	-
3D (All modes)	**39.514**	**0.987**	**0.021**	**+5.575 dB**

- **3D transform boosts** reconstruction performance (+5.575 dB over best 1D)
- Among 1D variants, **spectral transform (mode-3) outperforms** spatial transforms
- Suggests **spectral mode has stronger capacity** in capturing informative HSI structures
- **Cross-directional correlations** are critical for optimal recovery

Experiment: compare two latent tensor acquisition strategies

- **Learnable**: $\widehat{\mathcal{X}}_i$ as directly optimized variables
- **Generated**: $\widehat{\mathcal{X}}_i = f_i(\mathcal{X}_i^0)$ via neural networks

Results (Case 4, Pavia Centre):

Strategy	PSNR	SSIM	SAM	Parameters
Generated	36.823	0.980	0.025	$2\times$ network params
Learnable	**39.514**	**0.987**	**0.021**	Baseline

Advantages of Learnable Approach:

1. **Superior accuracy**: $+2.691$ dB PSNR improvement
2. **Parameter efficiency**: significantly fewer parameters (no generator networks)
3. **Optimization directness**: avoids generator network complexity

Finding: direct latent tensor optimization is more compact and effective than network-generated representations

Experiment: compare three initialization strategies for $\{\widehat{\mathcal{X}}_i\}_{i=1}^3$

- (a) **Random**: uniform distribution $\mathcal{U}(0,1)$
- (b) **Observed Image**: missing entries filled with zeros
- (c) **TNN Result**: conventional TNN method output

Results (Case 4, Pavia Centre):

Initialization	PSNR	SSIM	SAM	Performance Gap
$\mathcal{U}(0,1)$	39.078	0.986	0.022	-0.436 dB
Observed Image	39.302	0.987	0.022	-0.212 dB
TNN Result	**39.514**	**0.987**	**0.021**	Baseline

- **Low sensitivity**: all three strategies yield comparable performance (max gap 0.436 dB)
- **TNN initialization best**: provides more image information
- **Robustness advantage**: unlike 3DSTNN (sensitive to initialization), 3DeepRep ensures stable convergence

Experiment: Test expansion ratio k where $\widetilde{n}_i = kn_i$, $k \in \{0.5, 1, 2, 3, 4\}$

Results (Case 4, Pavia Centre):

k	PSNR	SSIM	SAM	Analysis
0.5	39.079	**0.987**	0.022	Insufficient information
1	**39.514**	**0.987**	**0.021**	**Optimal balance**
2	38.913	0.986	0.022	Compromised low-rankness
3	38.601	0.986	0.022	Further degradation
4	38.310	0.984	0.022	Excessive parameters

- **k=0.5**: Retains less image information for reconstruction
- **k=1**: Optimal balance between model fit and low-rank structure preservation
- **k>1**: Higher dimensions compromise low-rankness in specific modes, no performance gain

Design Principle: match latent space dimension to image dimension for optimal low-rank tensor representation

Hyperparameter Sensitivity Reveals Design Insights

1. θ: balance spatial (mode-1, mode-2) vs. spectral (mode-3) low-rankness
2. β: control overall low-rank regularization strength
3. γ: determine intermediate layer data fidelity importance

Sensitivity Analysis (Case 4, all datasets):

Parameter	Pavia Centre	Pavia Univ.	DC Mall	Sensitivity
θ	10^{-1}	10^{-1}	10^{0}	**High**
β	10^{-5}	10^{-5}	10^{-6}	**High**
γ	10^{-7}	10^{-8}	10^{-7}	**Low**

- Spatial-spectral balance is critical, varies by dataset characteristics
- Low-rank regularization is essential for performance
- Intermediate layer fidelity is relatively less critical

Practical Guidance: focus hyperparameter tuning on θ and β; γ can use default values

Experimental Validation Confirms 3DeepRep's Comprehensive Superiority

Quantitative Evidence:

- **Best PSNR** across 3 datasets, 4 scenarios, multiple missing rates
- **Largest gain**: +7.553 dB over CoNoT (Case 3, Pavia Centre)
- **Consistent improvement**: average +2.8 dB across all scenarios

Qualitative Evidence:

- **Sharpest details**: clearest features in all visual comparisons
- **Best color fidelity**: closest spectral tone to ground truth
- **Minimal error**: error maps closest to pure dark blue

Ablation Insights:

- **3D structure**: +5.575 dB over best 1D variant
- **Learnable latents**: +2.691 dB over generated latents
- **Initialization robust**: <0.5 dB variation across strategies
- **Optimal dimension**: k=1 balances fit and low-rankness

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Unified Framework

Common Foundation:

- Leraning-guided low-rank tensor representation
- Transform-based TNN framework
- Self-supervised optimization

Complementary Perspectives:

Dimension	CA-TNN	3DeepRep
Core Focus	Inter-channel differences	Inter-mode low-rankness
Weighting Strategy	Channel-level nuclear norm weighting	Direction-level reconstruction weighting
Transform Direction	Single (spectral mode)	Multi-directional (three modes)
Latent Representation	Single latent tensor	Three directional latent tensors
Aggregation	-	Neural network fusion (direction dim)

Complementarity of CA-TNN and 3DeepRep

CA-TNN Focus: **channel dimension**

- Fine-grained modeling of channel importance differences
- Assigns distinct weights to channel-wise nuclear norms
- Explicitly models channel characteristics and inter-channel relationships

3DeepRep Focus: **tensor mode dimension**

- Comprehensive modeling of directional low-rankness
- Enforces low-rankness along all three tensor modes
- Multi-directional deep transforms capture mode-specific correlations
- Three directional latent tensors with learnable aggregation

Future Directions

1. **Fusion of CA-TNN and 3DeepRep**: combine channel-aware weighting with multi-directional modeling
2. **Adaptive Hyperparameter Learning**: meta-learning or neural architecture search
3. **Extension to Other Degradation Types**: Gaussian noise, mixed noise-missing scenarios
4. **Multi-modal Fusion**: joint reconstruction of HSI with RGB, LiDAR, etc.
5. **Broader Applications**: medical imaging, video inpainting, 3D reconstruction

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CA-TNN Contributions:

- Identified and addressed channel weighting in existing WTNN methods
- Exponential FC layer for flexible weight range $(0, +\infty)$
- Proposed channel-aware TNN with learnable attention mechanism

3DeepRep Contributions:

- Identified and addressed direction inflexibility in existing transform-based TNN methods
- Learnable aggregation module for tensor fusion
- Direct latent tensor optimization for parameter efficiency and robustness
- Proposed multi-directional deep transform-based low-rank tensor representation

From Channel-Aware Weighting to Multi-Directional Deep Representation: learning-guided TNN is a powerful and flexible framework for HSI inpainting, with applications in remote sensing and other domains requiring high-fidelity tensor reconstruction.

Thank You!

Contact: Yunshan Li: lyunshan565@gmail.com
Wenwu Gong: gongww@sustech.edu.cn

Code: CA-TNN: `https://github.com/Kendradra/CA-TNN`
3DeepRep: `https://github.com/Kendradra/3DeepRep`

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